Date / /
Page

Information about the 2 algorithms

1) Genetic Algorithm

It is inspired by natural selection & genetics. It starts with a population of possible solutions & improves them through selection, crossover & mutation over multiple generations.

Application

- Engineering design & optimization
- Machine learning & feature selection
- Financial portfolio optimization
- Job scheduling & resource allocation
- Routing & planning problems

Advantages

- can solve complex optimization problems.
- Does not require ~~full~~ derivatives of the objective function
- works well with large & unexplored search spaces

Disadvantages

- can require many function evaluations
- Choosing population size & genetic parameters can be difficult
- May converge slowly.

Variants

origin - 1970

Latest - hybrid & multi-objective GA variants

Continue to be developed.

Date __/__/__
Page _____

2) Particle swarm optimization (PSO)

It is inspired by the social behaviours of birds & fish. A group of particles searches the solution space. Each particle adjusts its movement based on its personal best position & the global best position.

Application

- Neural - network optimization
- Power - system optimization
- controller & parameter tuning

Advantages

- Works well for continuous optimization
- Does not require gradient info
- Requires relatively few parameters.

Disadvantages

- can get trapped in local optima
- may lose diversity during optimization
- can converge prematurely.

Variants

Original - 1995

Robust - learning strategies, adaptive PSO, etc.

Date __/__/__
Page _____

3.) Ant colony optimization

It is inspired by ants searching for food. Artificial ants construct solutions by following pheromone trails & heuristic info.

Applications

- Travelling salesman problem
- scheduling
- path planning

Advantages

- very effective for routing & path-finding problems
- can adapt to changing environments
- suitable for combinatorial optimization

Disadvantages

- can be computationally expensive
- Premature convergence can affect performance
- usually slower than simpler methods for small problems

Notation

Original - 1990

Now - ant colony systems, Max-Min Ant systems

Date __/__/__
Page

4.) Cuckoo Search

It is inspired by the brood-parasitism behaviours of cuckoo birds. candidate solution are represented as nest. New solution are generated using Levy flights & poor nests are abandoned & replaced

Application

- Engineering design
- Machine learning
- image processing

Advantages

- Levy flights provide good exploration of the search space.
- Effective for continuous optimization.
- Suitable for complex nonlinear problems.

Disadvantages

- Performance depends on parameter settings
- can require many function evaluate
- Does not guarantee global optimum.

Version

original - 2009
latest - are develop adaptive, binary, hybrid & application - specific cuckoo search variant

Date __/__/__
Page _____

5) Grey wolf optimizer (GWO)

It is inspired by the social hierarchy & hunting behaviours of grey wolves. The best solution are represented by alpha, beta & delta, while grey. The remaining wolves towards promising area of the space.

Application

- controller design
- Robotics
- Engineering optimization

Advantages

- Simple mathematical structure
- Good balance between exploration & exploitation
- Does not require gradient info

Disadvantages

- can be trapped in local optima
- performance may degrade in population & complex
- Basic GWO may not perform well on any type of problem.

Variants
Exhibit - 2014
New - Alpha 2025-26 variant with mutation based GWO etc

Date __/__/__
Page _____

6) Parallel cellular Algs

It divides problem into cells arranged in a grid. Each cell interacts with neighbouring cells & updates its state based on local info. The operation can be performed in parallel

Application

- Simulation
- scheduling
- large scale optimization

Advantages

- Naturally suitable for parallel computing
- Local interaction can promote good exploration
- can handle large number of candidate solutions.

Disadvantages

- implementation can be more complicated
- Designing appropriate update rules can be difficult
- communication b/w cells can introduce overhead.

Version

Developed progressively. Modern development focus on parallel/distributed implementations, adaptive mutation algs & GPU-based approaches rather than one official new version

Date __/__/__
Page _____

7) Gene Expression Algo

It is inspired by gene expression in biological system. Solutions are represented as genetic sequences & are transformed & expressed into functional solutions. Selection, crossover & mutation are used to evolve better solutions

Applications

- Data mining
- complex system design
- Parameter optimization

Advantages

- can represent complex solutions
- can explore large search space
- useful for discovering complex relationship in data

Disadvantages

- can be computationally expensive
- requires careful parameter tuning
- may converge slowly

Variants

Earliest - 2001
Latest - continuous through hybrid, multi-objective & application - specific GEP variants, particularly in machine learning & genomic regression

CO1
24/9/26.